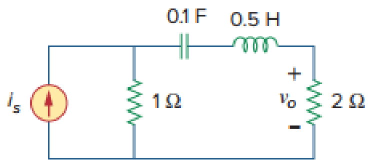
Chapter 9, Problem 51P

Problem

If the voltage *vo* across the 2-Ω resistor in the circuit of Fig. 9.58 is 90 cos 2*t* V, obtain *is*.

Figure 9.58:



Step-by-step solution

Step 1 of 6

Refer to Figure 9.58 in the text book.

Consider the value of the voltage $v_o(t)$.

$$v_o(t) = 90\cos 2t \text{ V}$$

Represent in phasor form.

$$\mathbf{V}_o = 90\angle 0° \text{ V}$$

Therefore, the value of the voltage in phasor form is $90\angle 0° \text{ V}$.

Step 2 of 6

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute 2 rad/s for ω and 0.5 H for L.

$$\begin{aligned} X_L &= j(2)(0.5) \\ &= j1 \\ &= 1\angle 90° \ \Omega \end{aligned}$$

Therefore, the value of the reactance X_L is $1\angle 90° \ \Omega$.

Step 3 of 6

Calculate the value of the reactance due to capacitance.

$$X_C = \frac{1}{j\omega C}$$

Substitute 2 rad/s for ω and 0.1 F for C.

$$\begin{aligned} X_C &= \frac{1}{j(2)(0.1)} \\ &= \frac{1}{j0.2} \\ &= 5\angle -90° \ \Omega \end{aligned}$$

Therefore, the value of the reactance X_C is $5\angle -90° \ \Omega$.

Step 4 of 6

Draw the circuit diagram in phasor representation.

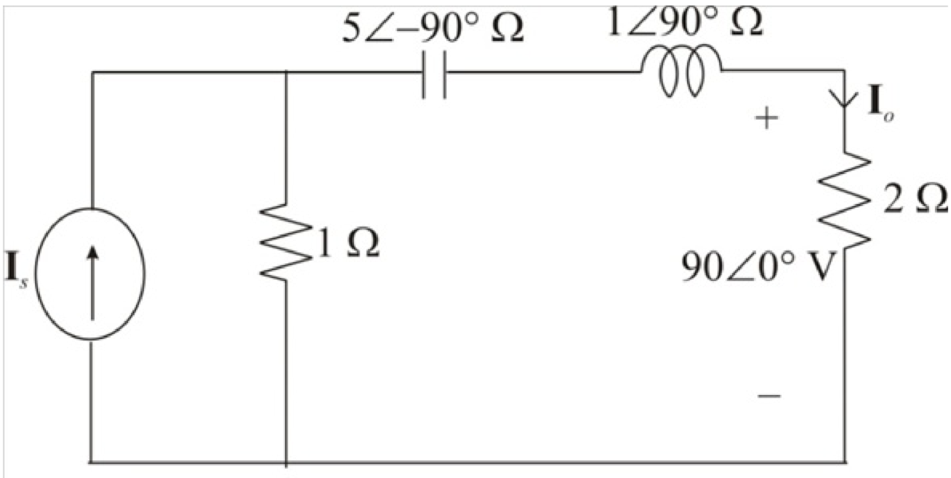


Figure 1

Step 5 of 6

Calculate the value of the current $\mathbf{I}_o$.

$$\mathbf{I}_o = \frac{\mathbf{V}_o}{2}$$

Substitute $90\angle 0°\ \text{V}$ for $\mathbf{V}_o$.

$$\begin{aligned}\mathbf{I}_o &= \frac{90\angle 0°}{2}\\ &= 45\angle 0°\ \text{A}\end{aligned}$$

Therefore, the value of the current $\mathbf{I}_o$ is $45\angle 0°\ \text{A}$.

Step 6 of 6

Write the expression for the current $\mathbf{I}_o$ using current divider rule.

$$\mathbf{I}_o = \mathbf{I}_s\left(\frac{1}{1+(1\angle 90° + 5\angle -90° + 2)}\right)$$

Substitute $45\angle 0°\ \text{A}$ for $\mathbf{I}_o$.

$$45\angle 0° = \mathbf{I}_s\left(\frac{1}{5\angle -53.13°}\right)$$

$$45\angle 0° = \mathbf{I}_s(0.2\angle 53.13°)$$

$$\mathbf{I}_s = \frac{45\angle 0°}{0.2\angle 53.13°}$$

$$\mathbf{I}_s = 225\angle -53.13°\ \text{A}$$

Represent in time domain.

$$i_s(t) = 225\cos(2t - 53.13°)\ \text{A}$$

Therefore, the value of the current $i_s(t)$ is $\boxed{225\cos(2t - 53.13°)\ \text{A}}$.